

Tech & Teach

Digital Chip Design Program

RTL Design

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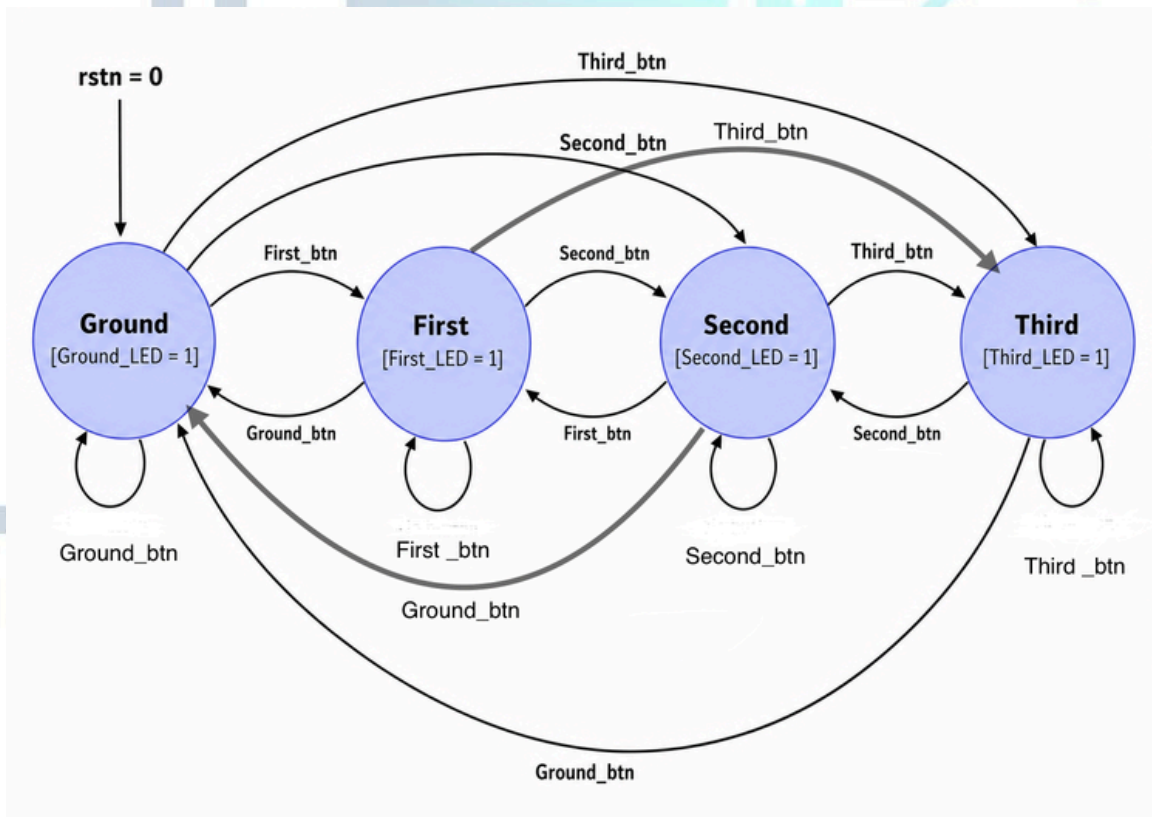
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Controller – Part 2

Version 1.0

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Task #13.1



Task #13.1

```
module fsm_1(  
    input logic Ground_btn,  
    input logic First_btn,  
    input logic Second_btn,  
    input logic Third_btn,  
    input logic clk,  
    input logic rstn,  
  
    output logic Ground_LED,  
    output logic First_LED,  
    output logic Second_LED,  
    output logic Third_LED  
);  
  
enum logic [2:0] {  
    Ground,  
    First,  
    Second,  
    Third,  
    unused = 3'bxxx  
} current_state, next_state;
```

```
// Output Logic  
  
always @(current_state)  
case (current_state)  
  
    Ground: begin  
        Ground_LED = 1;  
        First_LED = 0;  
        Second_LED = 0;  
        Third_LED = 0;  
    end  
  
    First: begin  
        Ground_LED = 0;  
        First_LED = 1;  
        Second_LED = 0;  
        Third_LED = 0;  
    end  
  
    Second: begin  
        Ground_LED = 0;  
        First_LED = 0;  
        Second_LED = 1;  
        Third_LED = 0;  
    end  
  
    Third: begin  
        Ground_LED = 0;  
        First_LED = 0;  
        Second_LED = 0;  
        Third_LED = 1;  
    end  
  
    default: begin  
        Ground_LED = 1'bx;  
        First_LED = 1'bx;  
        Second_LED = 1'bx;  
        Third_LED = 1'bx;  
    end  
  
endcase
```

Task #13.1

```
// Next State Logic
always @(current_state, Ground_btn,
  First_btn, Second_btn, Third_btn)
case(current_state)
Ground: begin
  if (First_btn)
    next_state = First;
  else if (Second_btn)
    next_state = Second;
  else if (Third_btn)
    next_state = Third;
  else
    next_state = Ground;
end
First: begin
  if (Ground_btn)
    next_state = Ground;
  else if (Second_btn)
    next_state = Second;
  else if (Third_btn)
    next_state = Third;
  else
    next_state = First;
end
Second: begin
  if (Ground_btn)
    next_state = Ground;
  else if (First_btn)
    next_state = First;
  else if (Third_btn)
    next_state = Third;
  else
    next_state = Second;
end
Third: begin
  if (Ground_btn)
    next_state = Ground;
  else if (First_btn)
    next_state = First;
  else if (Second_btn)
    next_state = Second;
  else
    next_state = Third;
end
default:
  next_state = unused;
endcase
```

```
// State Register

always @(posedge clk or negedge rstn)
begin
  if (!rstn)
    current_state <= Ground;
  else
    current_state <= next_state;
end

endmodule
```

Task #13.1

```
`timescale 1ns/1ps
```

```
module fsm_1_tb;
```

```
logic Ground_btn;  
logic First_btn;  
logic Second_btn;  
logic Third_btn;  
logic clk;  
logic rstn;
```

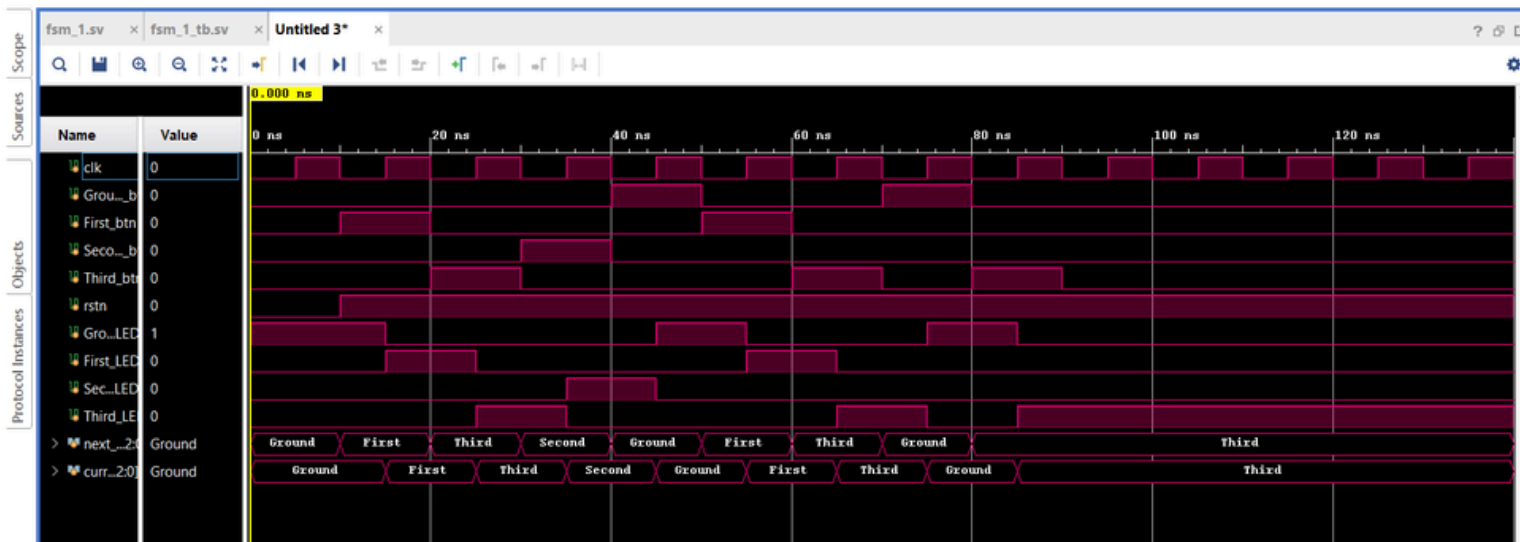
```
logic Ground_LED;  
logic First_LED;  
logic Second_LED;  
logic Third_LED;
```

```
fsm_1 dut(  
    .Ground_btn(Ground_btn),  
    .First_btn(First_btn),  
    .Second_btn(Second_btn),  
    .Third_btn(Third_btn),  
    .clk(clk),  
    .rstn(rstn),  
    .Ground_LED(Ground_LED),  
    .First_LED(First_LED),  
    .Second_LED(Second_LED),  
    .Third_LED(Third_LED)  
);
```

```
always begin  
    #5 clk = ~clk;  
end  
initial begin  
    clk=0;  
    rstn = 0;  
    Ground_btn = 0;  
    First_btn = 0;  
    Second_btn = 0;  
    Third_btn = 0;  
    #10;  
    rstn = 1;  
    First_btn = 1;  
    #10;  
    First_btn = 0;
```

```
    Third_btn = 1;  
    #10;  
    Third_btn = 0;  
  
    Second_btn = 1;  
    #10;  
    Second_btn = 0;  
    Ground_btn = 1;  
    #10;  
    Ground_btn = 0;
```

```
    First_btn = 1;  
    #10;  
    First_btn = 0;  
    Third_btn = 1;  
    #10;  
    Third_btn = 0;  
    Ground_btn = 1;  
    #10;  
    Ground_btn = 0;  
    Third_btn = 1;  
    #10;  
    Third_btn = 0;  
    #50;  
    $finish;  
end  
endmodule
```



Work distribution

Section	Almas	Atheer	Mjd	Maha	Jood
Report	✓	✓	✓	✓	✓
Task13.1	✓	✓	✓	✓	✓
Task13.2	✓	✓	✓	✓	✓
Task13.3	✓	✓	✓	✓	✓

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